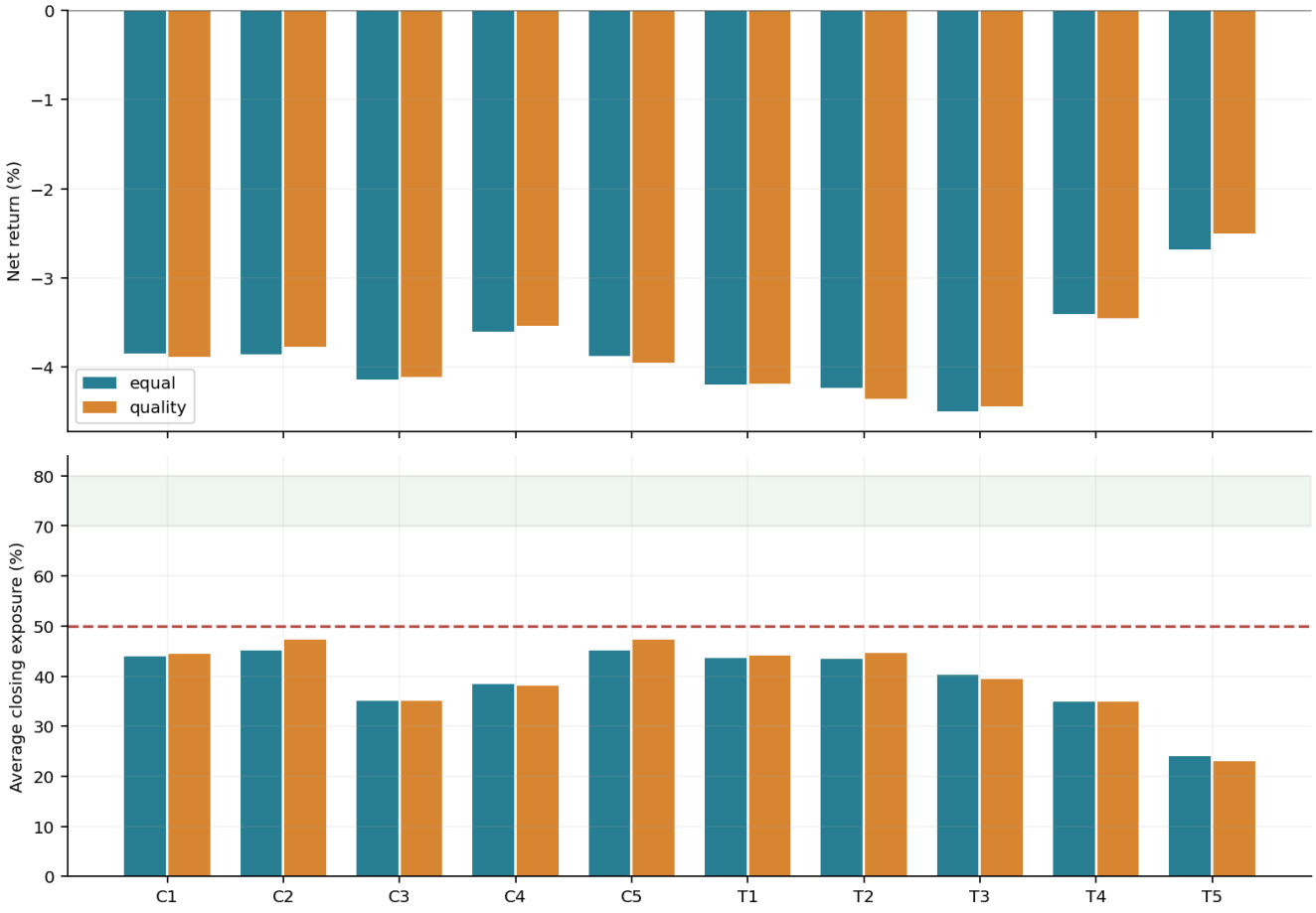
Phase 5 · Portfolio and execution research

20 paired simulations · 8 sensitivity runs · 48 supplied constituents · 16 September development sessions

Common criteria: 0/20 pass. Best Alpha: 0/20 pass. No independent validation or candidate promotion.

Equal and quality sizing use identical daily stock filters and execution rules. Quality is a research formula, not verified future profitability.



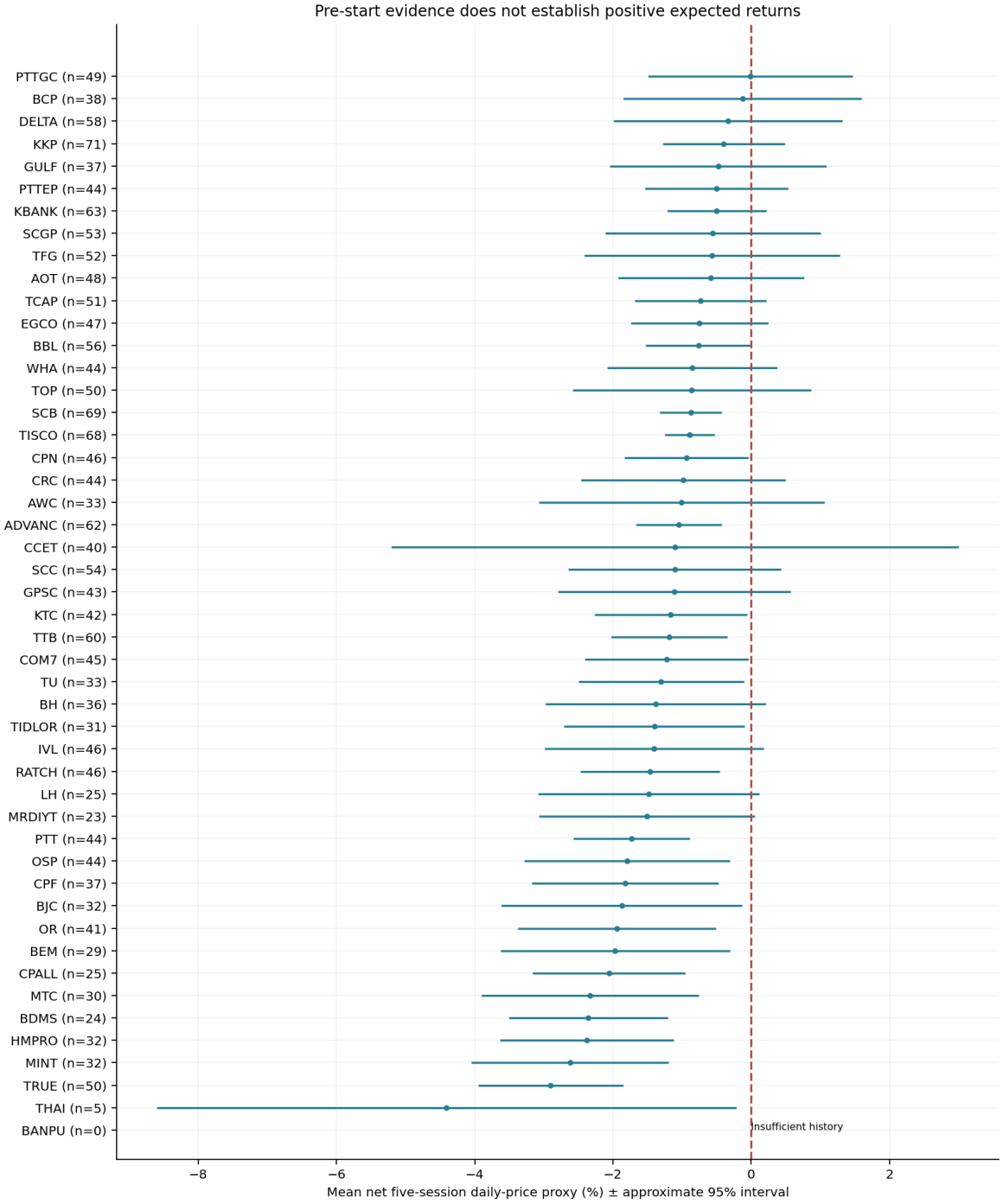
Measured base results

Run	Return %	DD %	Exp %	Trips	Payoff	Common
C1-equal	-3.85	4.10	43.9	445	-633.7	fail
C1-quality	-3.89	4.01	44.5	393	-803.8	fail
C2-equal	-3.86	4.03	45.1	438	-705.3	fail
C2-quality	-3.78	3.76	47.3	360	-898.1	fail
C3-equal	-4.14	4.64	35.1	341	-1,044.7	fail
C3-quality	-4.11	4.62	35.1	355	-989.1	fail
C4-equal	-3.61	3.42	38.4	535	-560.0	fail
C4-quality	-3.53	3.31	38.2	531	-556.0	fail
C5-equal	-3.88	4.03	45.2	421	-750.3	fail
C5-quality	-3.95	3.94	47.3	375	-898.4	fail
T1-equal	-4.19	4.46	43.6	457	-689.2	fail
T1-quality	-4.18	4.28	44.2	446	-742.8	fail
T2-equal	-4.23	4.38	43.5	431	-814.3	fail
T2-quality	-4.36	4.39	44.6	415	-869.5	fail
T3-equal	-4.50	4.87	40.3	411	-815.2	fail
T3-quality	-4.45	4.83	39.5	446	-770.9	fail
T4-equal	-3.41	3.23	35.0	507	-508.3	fail
T4-quality	-3.45	3.26	34.9	504	-536.6	fail
T5-equal	-2.68	2.62	24.0	317	-721.6	fail
T5-quality	-2.50	2.44	23.0	335	-630.8	fail

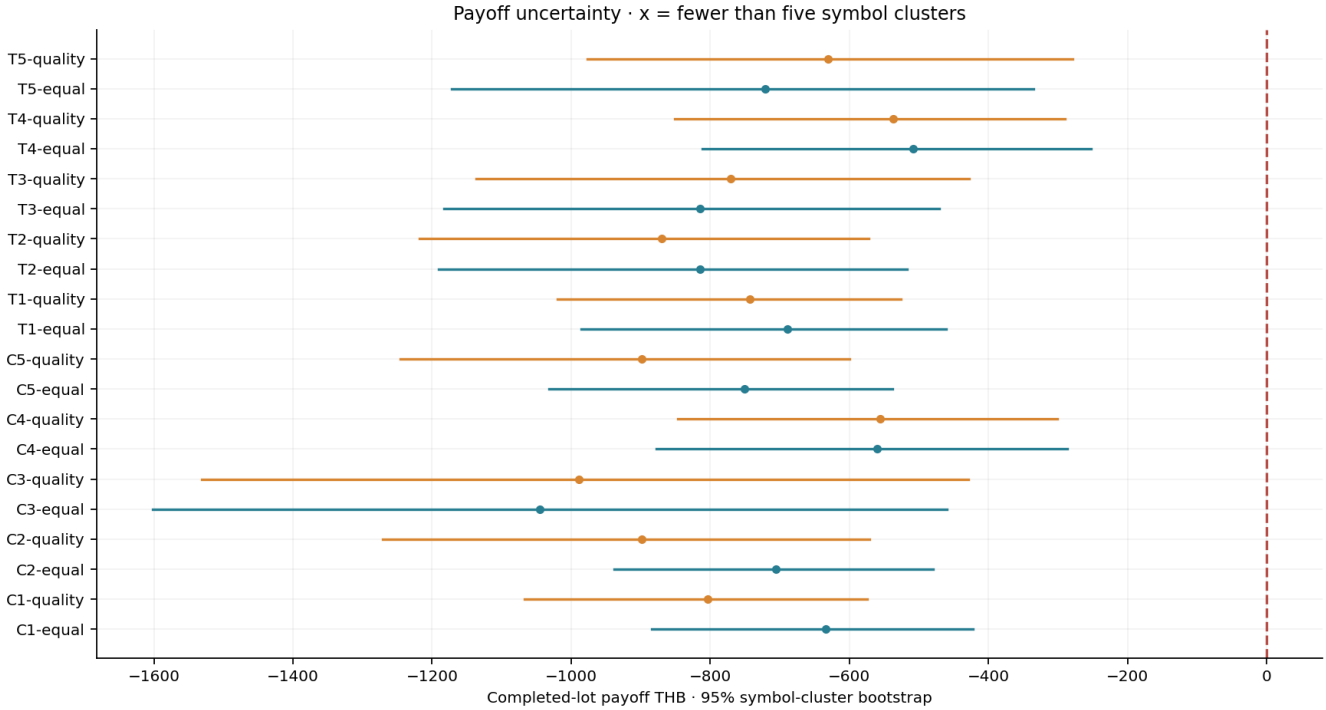
Round trips are completed FIFO buy lots. User target >5; common rule >=5. N/A means undefined payoff, not zero. The CSV also reports position episodes, excess return, initial-capital drawdown, open losses and costs.

Historical support and its limits

No stock had positive after-cost mean in the pre-start five-session proxy. The positive-edge veto produced no trades and was rejected as infeasible. The paired runs test trend and risk hypotheses with this uncertainty disclosed.



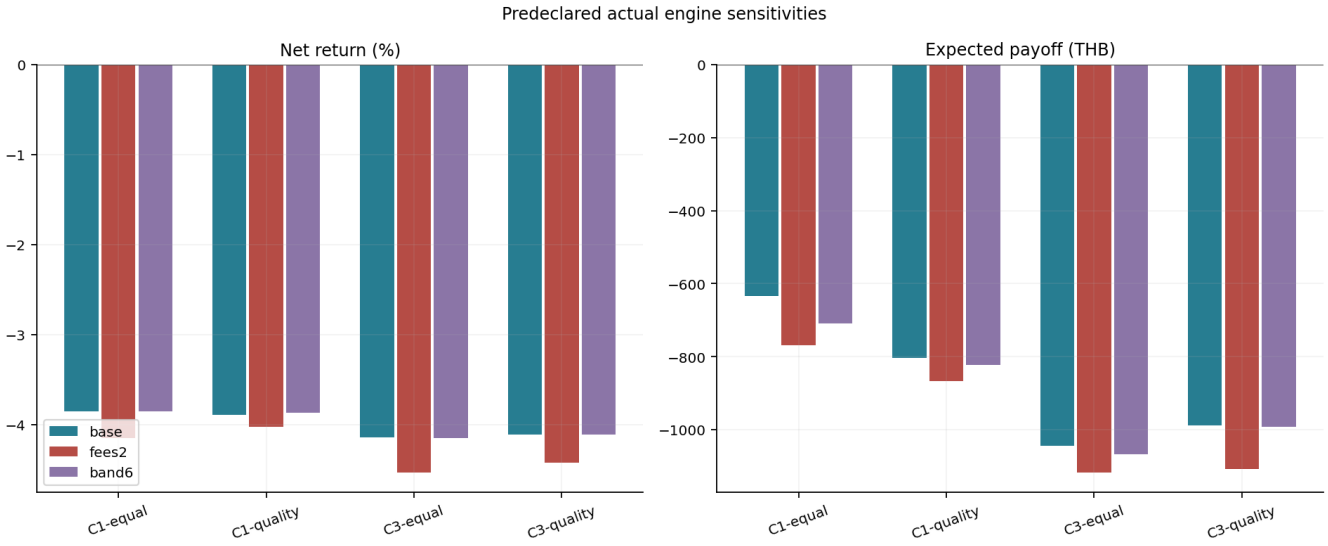
Payoff uncertainty



2,000 symbol-cluster resamples. Fewer than five symbol clusters: x, no interval. Market-wide dependence and repeated development decisions remain. More FIFO lots do not imply more independent profitable trades.

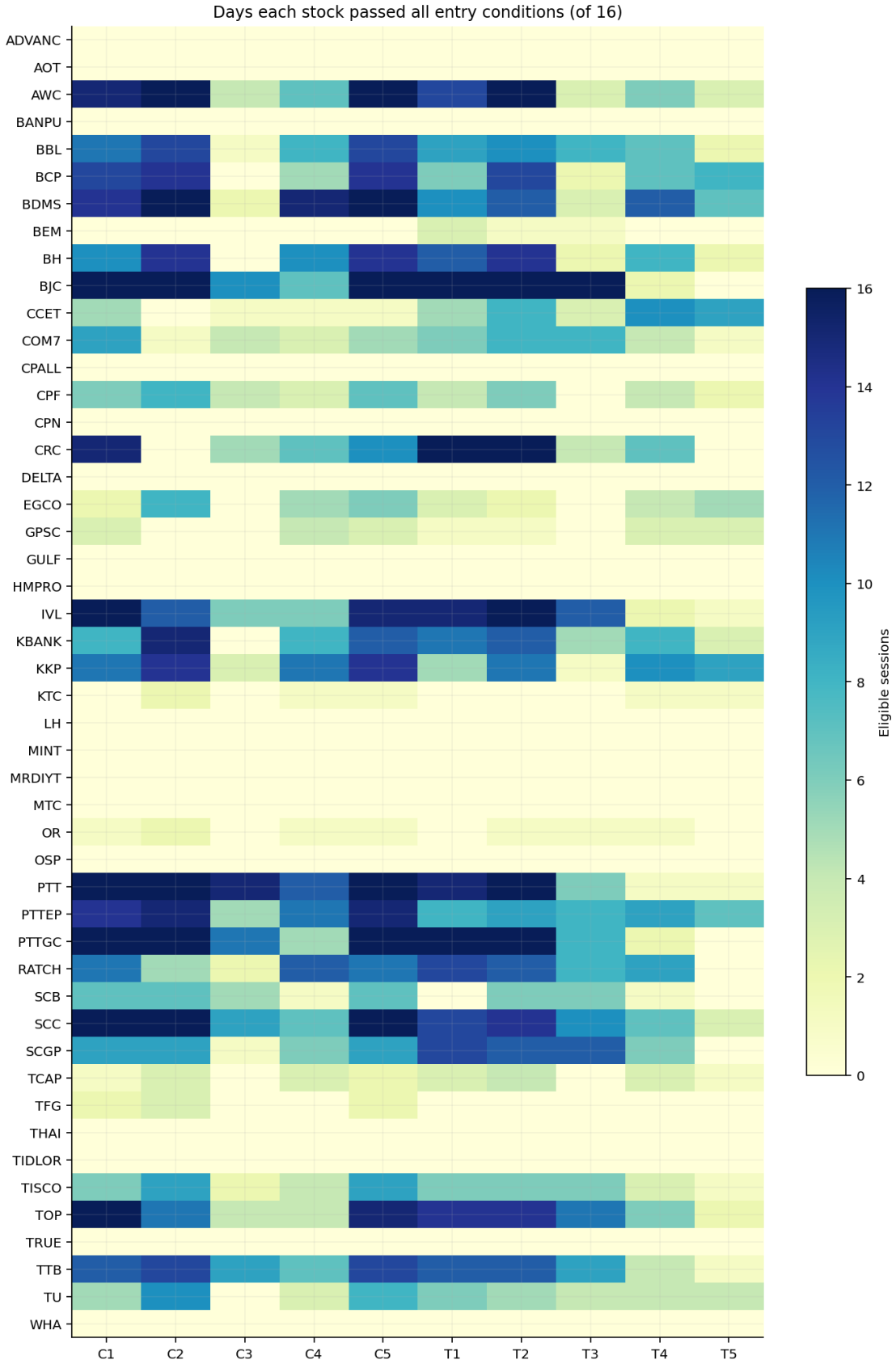
Stock control: past-only data, liquidity, volatility, long trend and model-specific predicates. Risk control: position/sector limits, 20% cash reserve, recovering drawdown guard, stop exits, daily buy budgets and actual residual IOC fills.

Sensitivity to actual costs and turnover

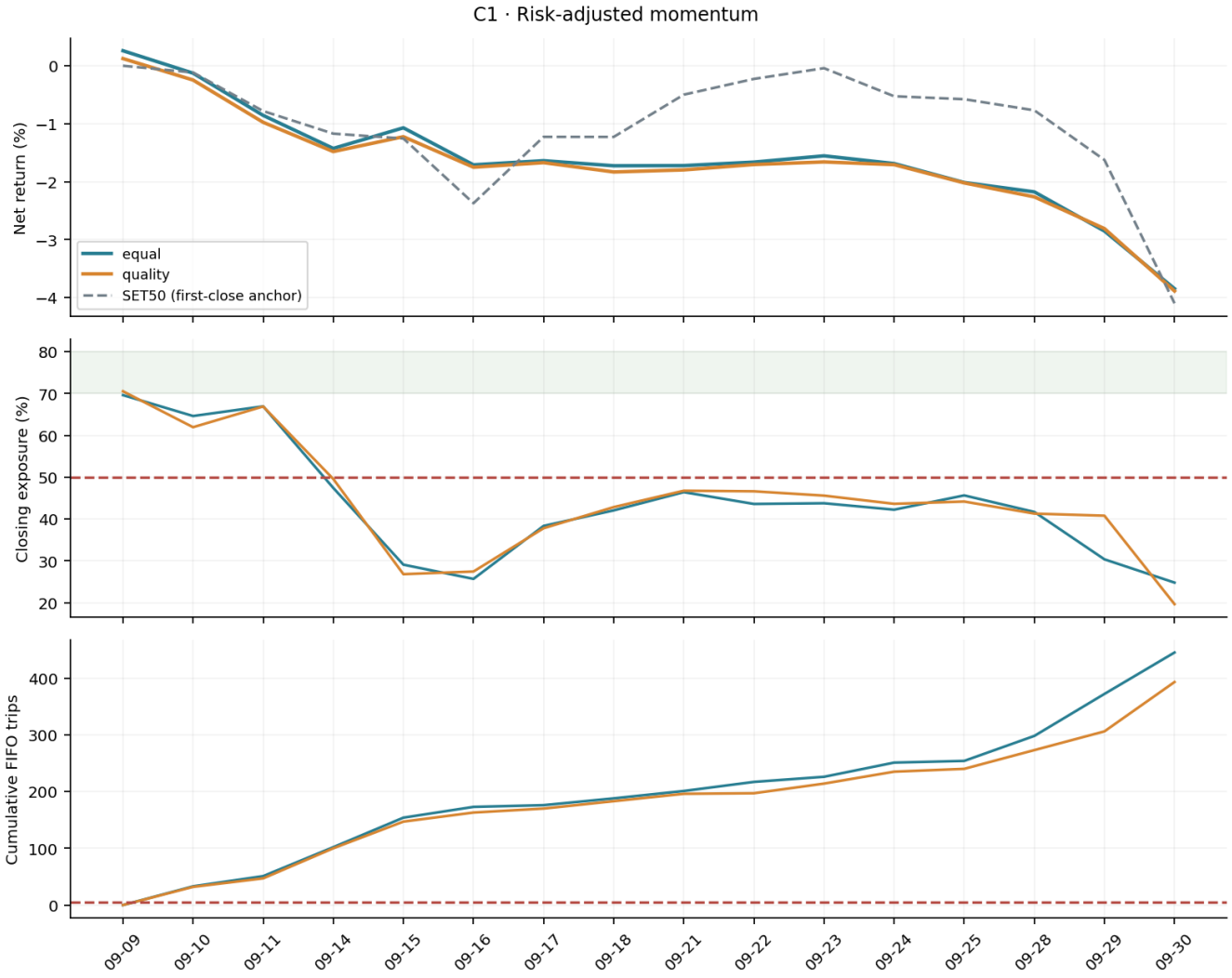


Double-commission and 0.6%-NAV-band variants were declared for C1/C3 before full results. Every variant reruns the official matching engine and reconciles resulting fills. No package files were edited.

All available stocks are assessed



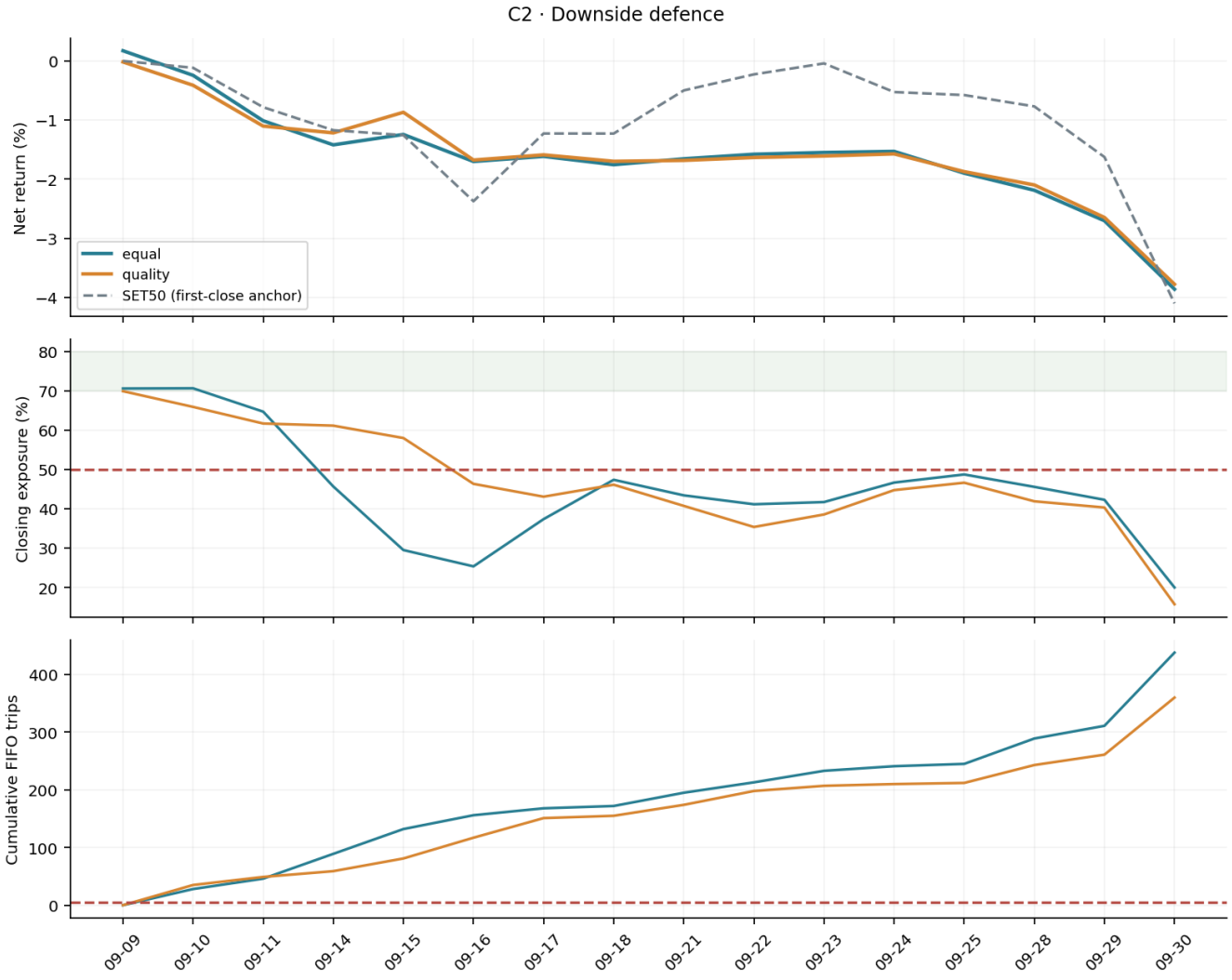
C1 · Risk-adjusted momentum



equal: return -3.847%; exposure 43.893%; 445 FIFO trips; 41 completed position episodes; payoff THB -633.73. Common: fail; Best Alpha: fail.

quality: return -3.888%; exposure 44.522%; 393 FIFO trips; 42 completed position episodes; payoff THB -803.80. Common: fail; Best Alpha: fail.

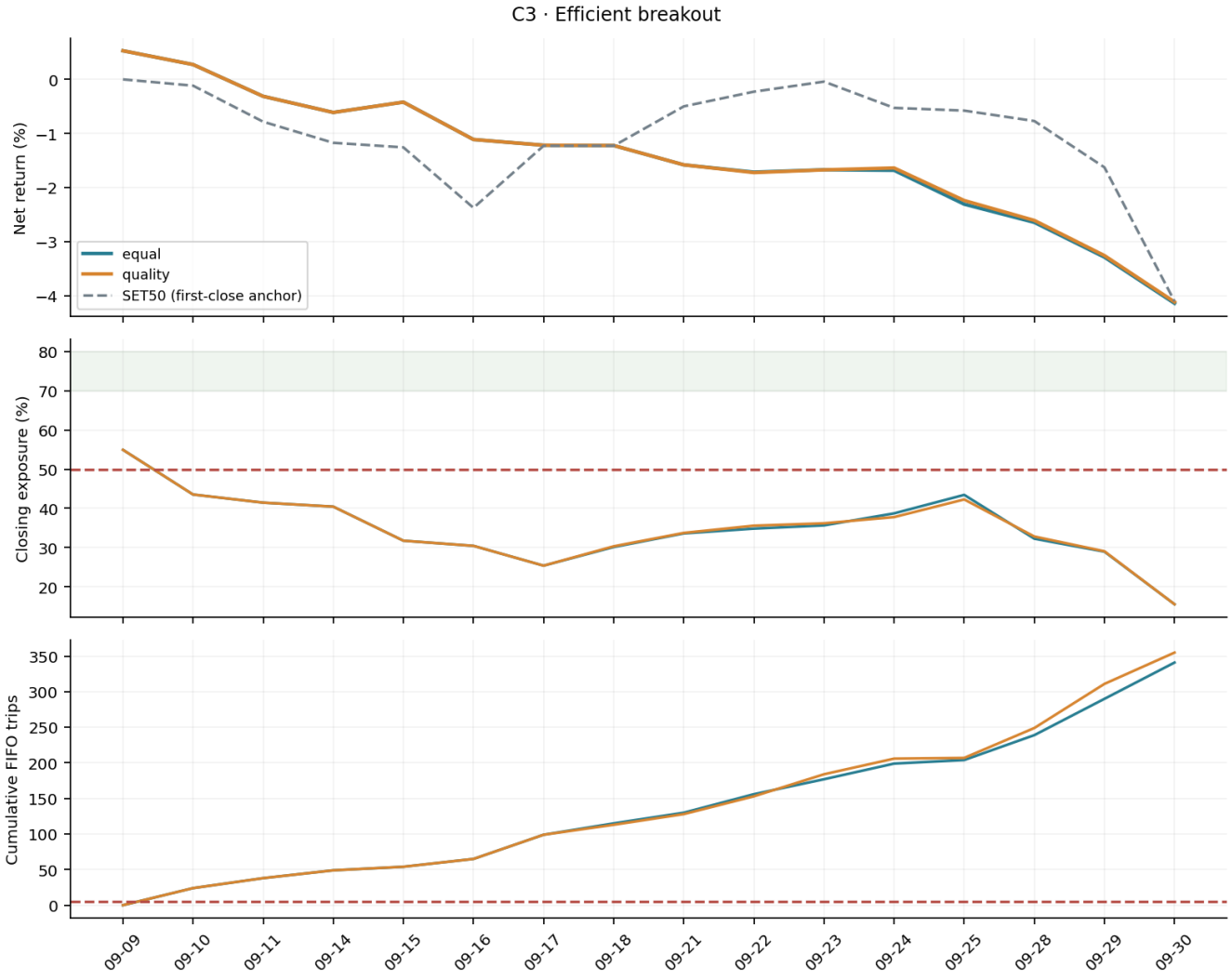
C2 · Downside defence



equal: return -3.861%; exposure 45.077%; 438 FIFO trips; 40 completed position episodes; payoff THB -705.29. Common: fail; Best Alpha: fail.

quality: return -3.777%; exposure 47.292%; 360 FIFO trips; 43 completed position episodes; payoff THB -898.10. Common: fail; Best Alpha: fail.

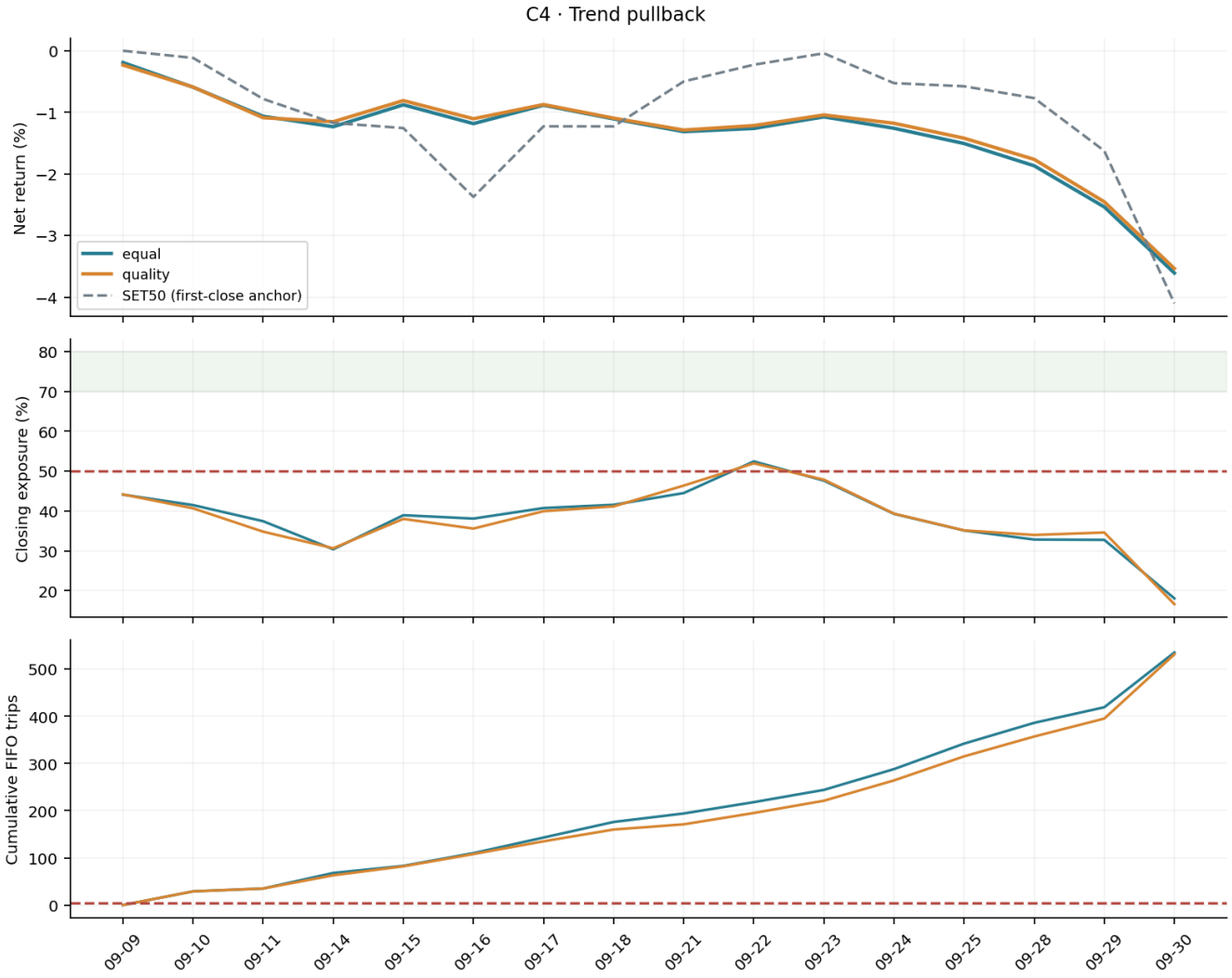
C3 · Efficient breakout



equal: return -4.137%; exposure 35.069%; 341 FIFO trips; 30 completed position episodes; payoff THB -1,044.74. Common: fail; Best Alpha: fail.

quality: return -4.111%; exposure 35.079%; 355 FIFO trips; 30 completed position episodes; payoff THB -989.09. Common: fail; Best Alpha: fail.

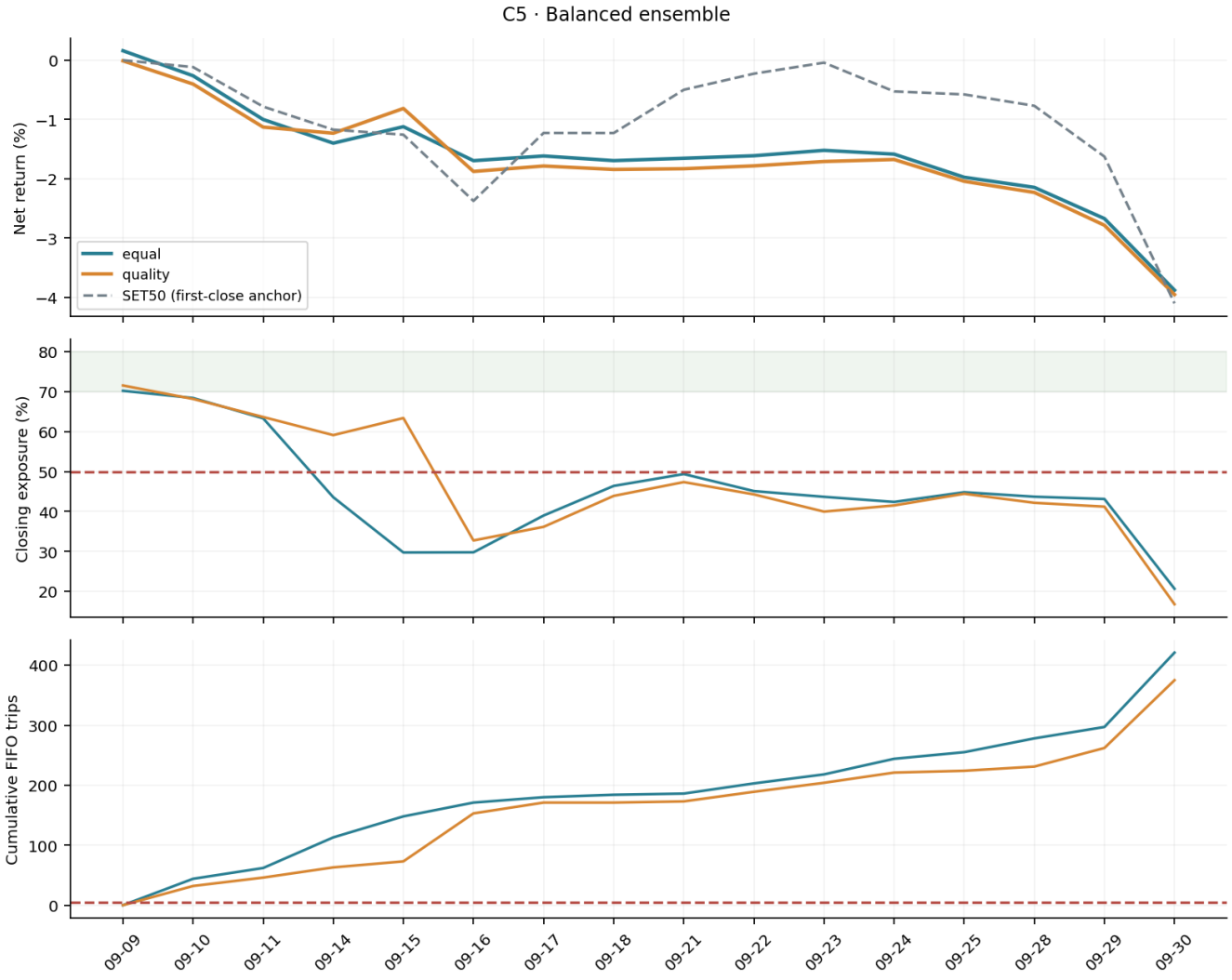
C4 · Trend pullback



equal: return -3.608%; exposure 38.439%; 535 FIFO trips; 56 completed position episodes; payoff THB -559.95. Common: fail; Best Alpha: fail.

quality: return -3.535%; exposure 38.159%; 531 FIFO trips; 56 completed position episodes; payoff THB -556.03. Common: fail; Best Alpha: fail.

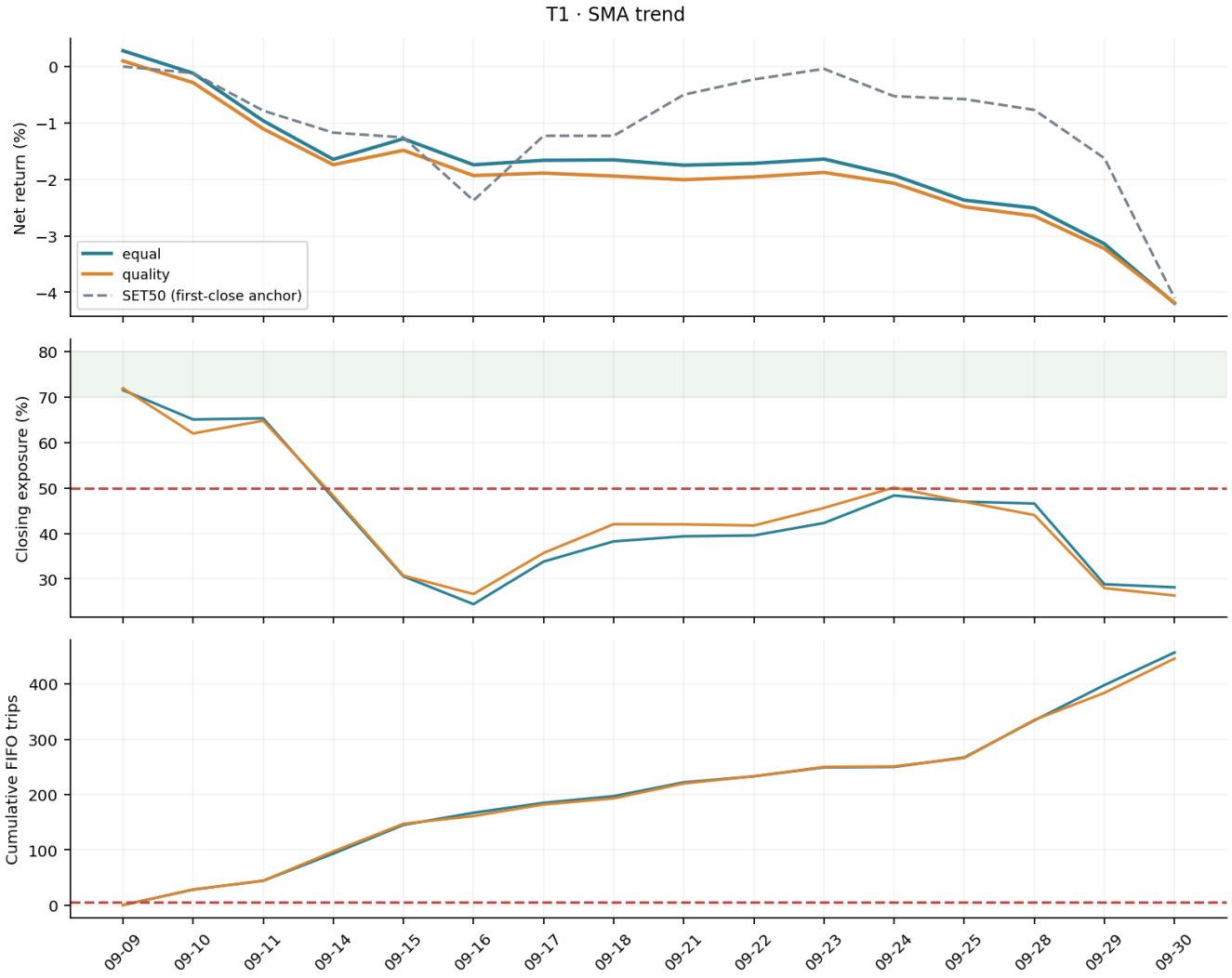
C5 · Balanced ensemble



equal: return -3.877%; exposure 45.195%; 421 FIFO trips; 44 completed position episodes; payoff THB -750.25. Common: fail; Best Alpha: fail.

quality: return -3.952%; exposure 47.267%; 375 FIFO trips; 45 completed position episodes; payoff THB -898.40. Common: fail; Best Alpha: fail.

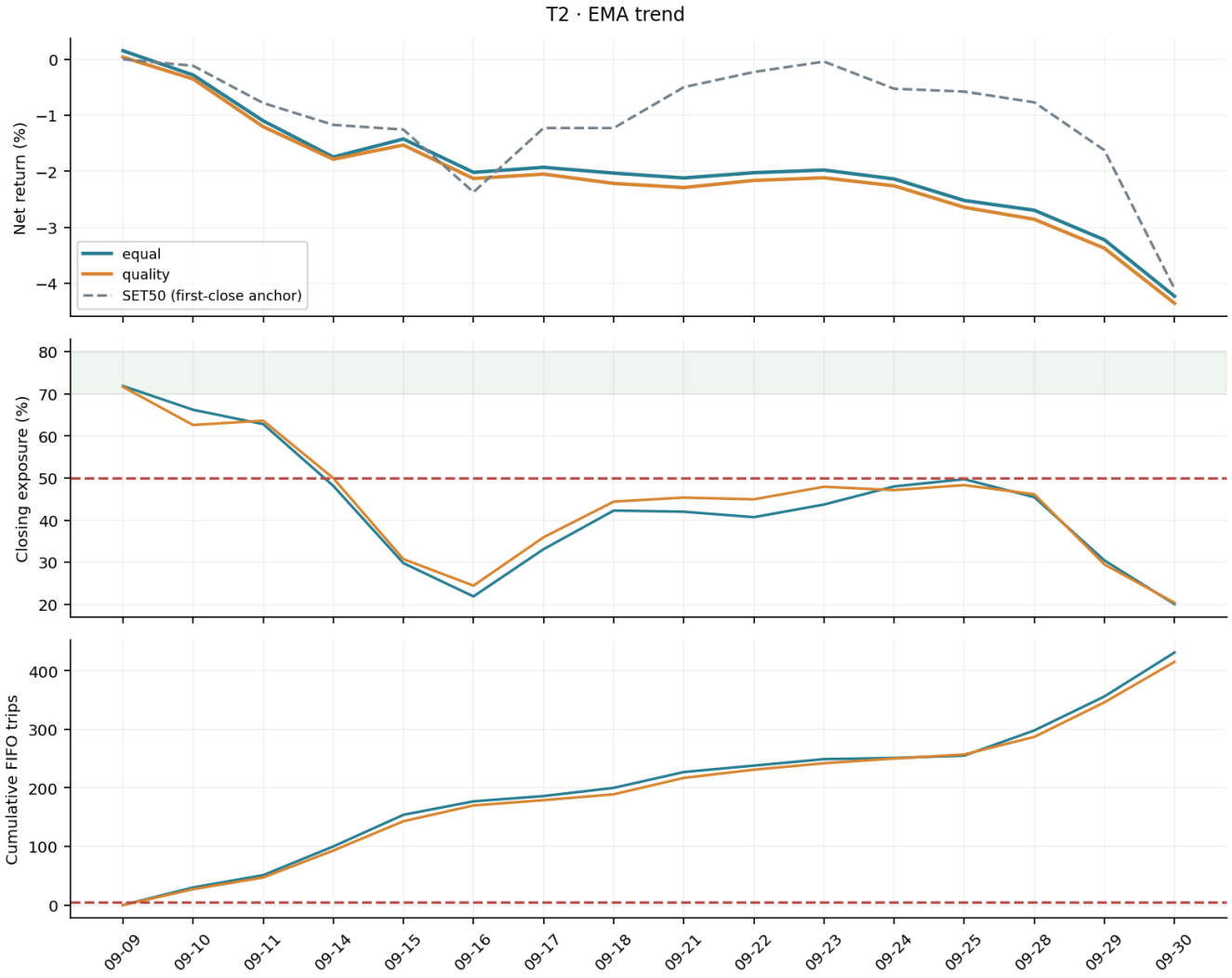
T1 · SMA trend



equal: return -4.195%; exposure 43.574%; 457 FIFO trips; 41 completed position episodes; payoff THB -689.16. Common: fail; Best Alpha: fail.

quality: return -4.184%; exposure 44.201%; 446 FIFO trips; 40 completed position episodes; payoff THB -742.83. Common: fail; Best Alpha: fail.

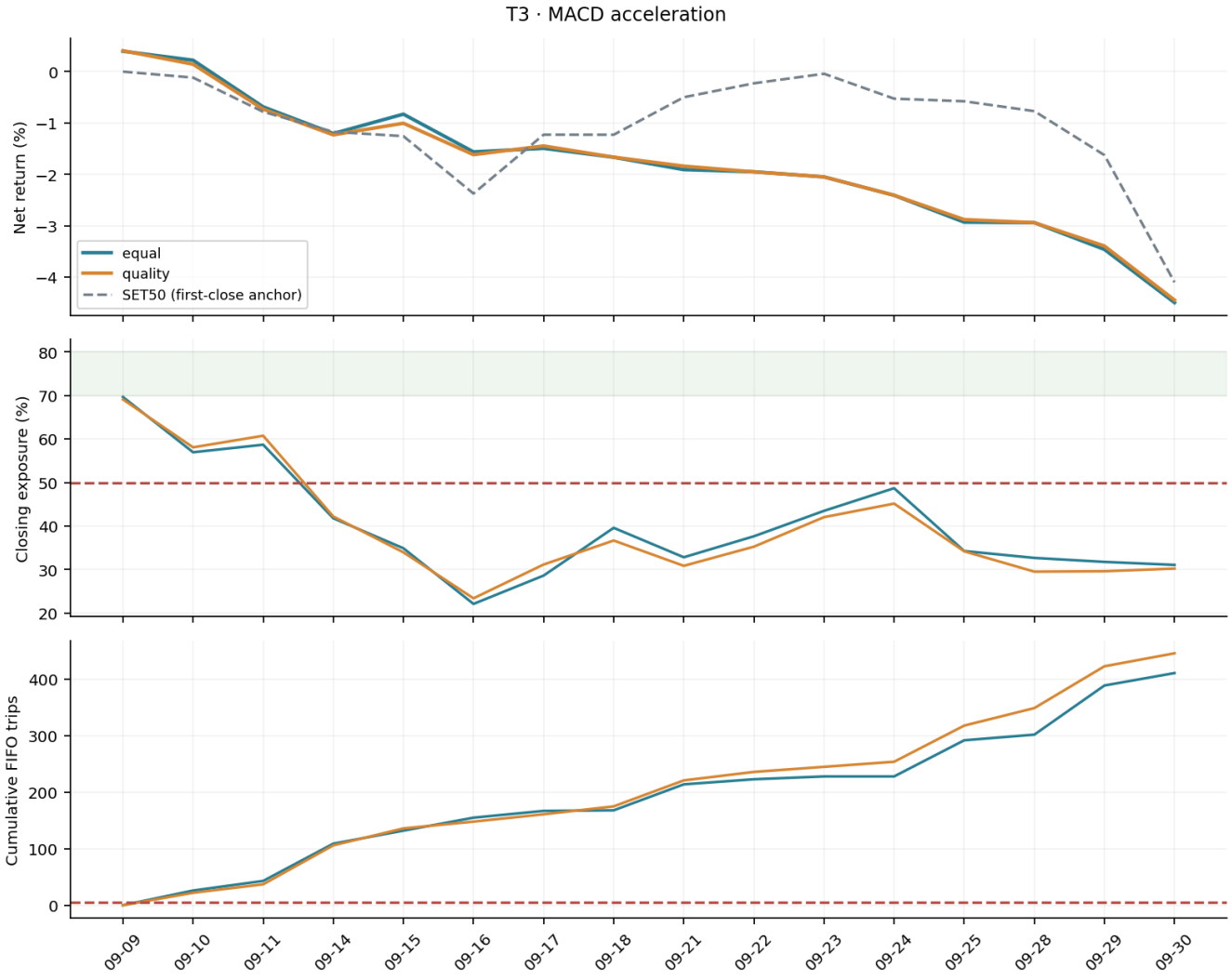
T2 · EMA trend



equal: return -4.231%; exposure 43.501%; 431 FIFO trips; 47 completed position episodes; payoff THB -814.30. Common: fail; Best Alpha: fail.

quality: return -4.357%; exposure 44.558%; 415 FIFO trips; 46 completed position episodes; payoff THB -869.53. Common: fail; Best Alpha: fail.

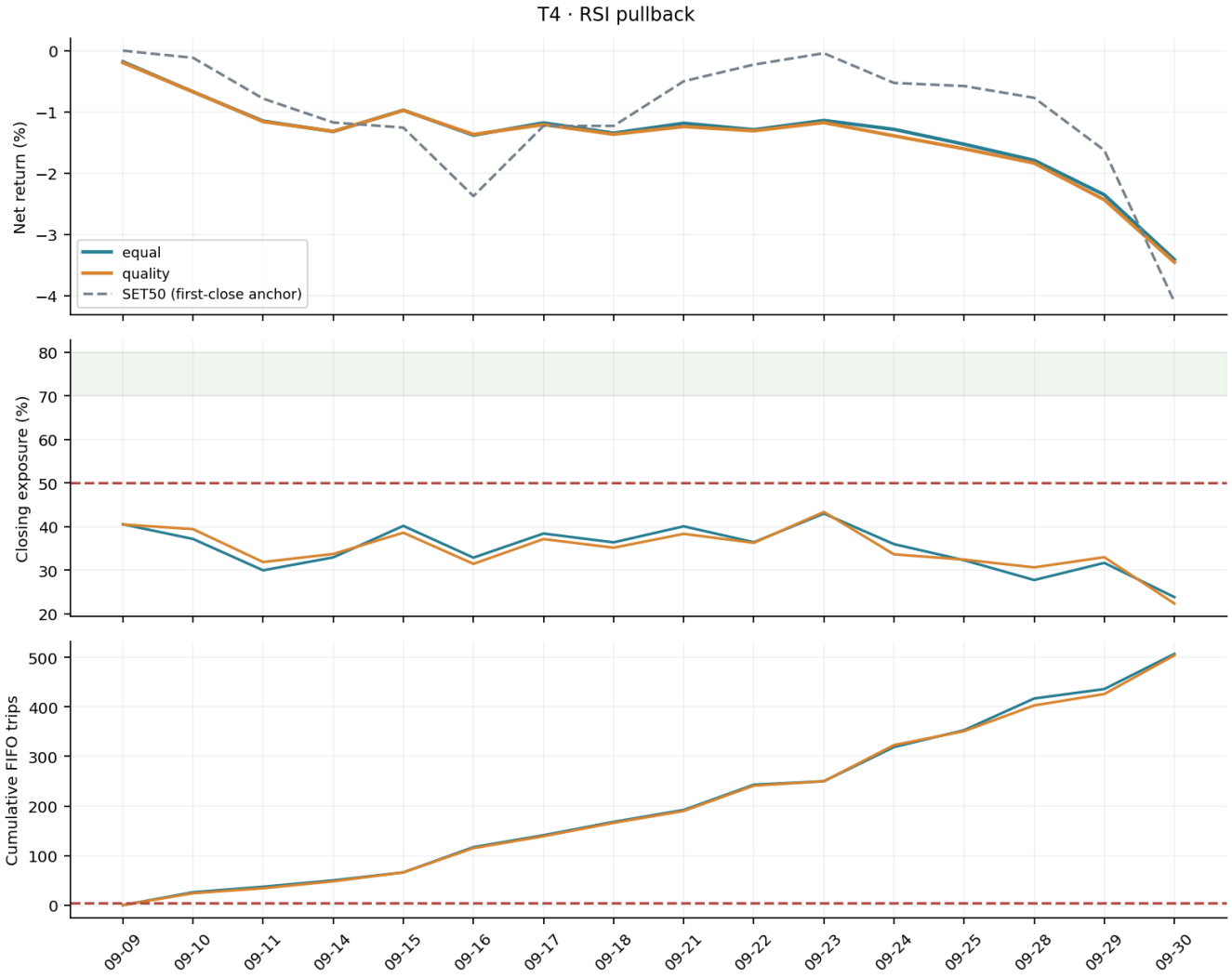
T3 · MACD acceleration



equal: return -4.495%; exposure 40.296%; 411 FIFO trips; 28 completed position episodes; payoff THB -815.23. Common: fail; Best Alpha: fail.

quality: return -4.446%; exposure 39.512%; 446 FIFO trips; 28 completed position episodes; payoff THB -770.86. Common: fail; Best Alpha: fail.

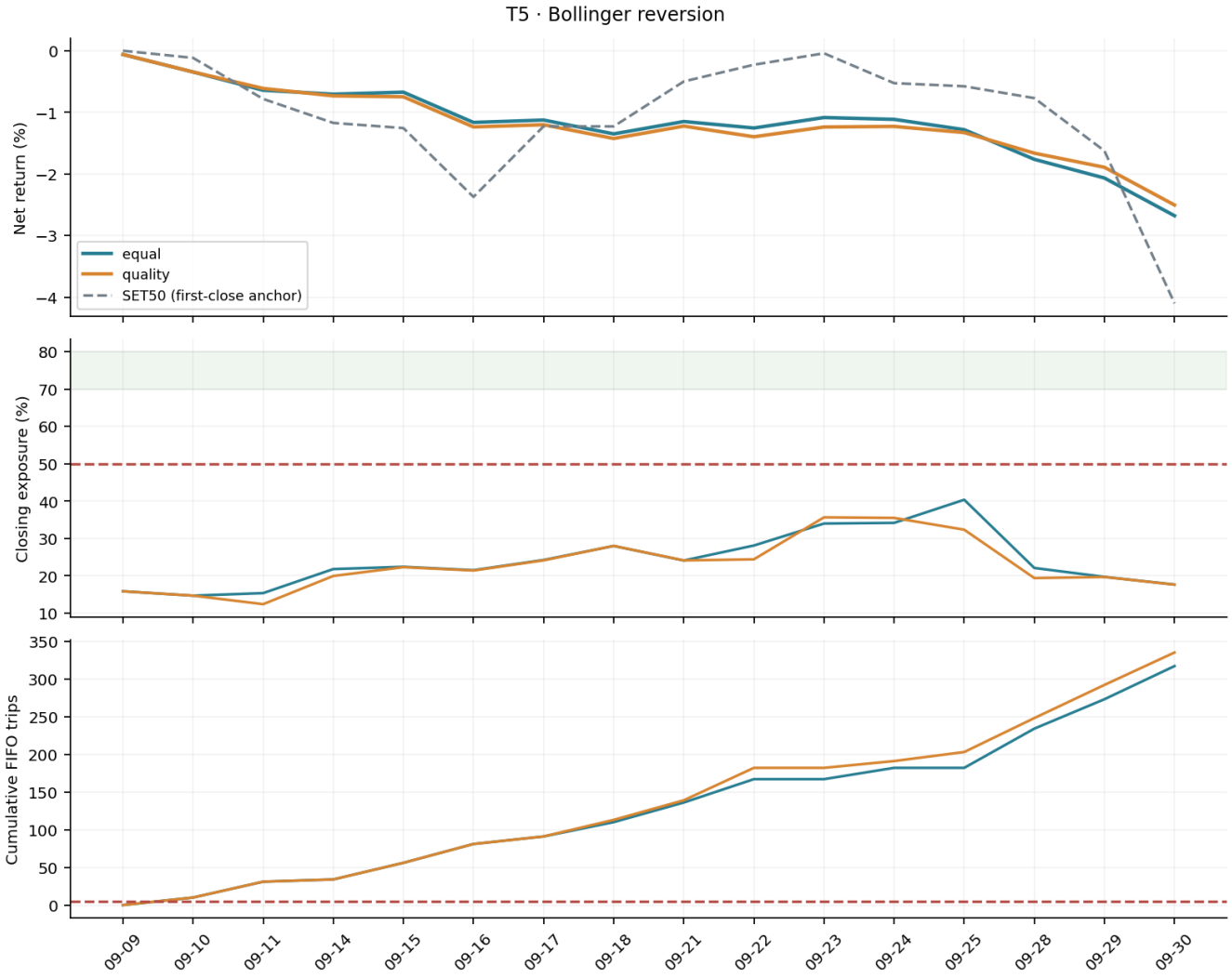
T4 · RSI pullback



equal: return -3.407%; exposure 34.991%; 507 FIFO trips; 47 completed position episodes; payoff THB -508.33. Common: fail; Best Alpha: fail.

quality: return -3.453%; exposure 34.894%; 504 FIFO trips; 47 completed position episodes; payoff THB -536.63. Common: fail; Best Alpha: fail.

T5 · Bollinger reversion



equal: return -2.678%; exposure 24.010%; 317 FIFO trips; 29 completed position episodes; payoff THB -721.55. Common: fail; Best Alpha: fail.

quality: return -2.504%; exposure 22.975%; 335 FIFO trips; 30 completed position episodes; payoff THB -630.77. Common: fail; Best Alpha: fail.

Verification and phase status

108 tests pass; 448 strategy-days reconcile; nine deterministic artifacts match a fresh C1-quality repeat.

4 queued cases failed package preflight before trading during a preparation-path check. The original package checksum was restored and verified; separate successful retries and the original failed logs are retained. A safe preparation entrypoint now rejects output inside the simulator directory.

Original Phase 3/4 strategy and declarations remain intact. Reports include signed source/config hashes, stock conditions and fill-level accounting. The model has no future tick or final-stock-P&L input.

Engineering and development experiments are complete. The Phase 5 improvement gate and Phase 4 independent validation remain open: no candidate is promoted. See docs/to-do/phase-6.md for the remaining research and exact-artifact rehearsal path.

Limits: static 48-name membership; reused September data; sampled stops and asynchronous cross-symbol marks; process-local intraday state; organizer submission access and identity not confirmed. Historical win fractions and payoff do not establish a high probability of future profit.